

ELITE MATHEMATICS

Pearson Edexcel International GCSE

# Linear 4MA1

## Adaptive Classified

OL-T49 · Sine Rule, Cosine Rule and Area of a Triangle

### Questions

Student book · questions with writing space

■ 13 pages

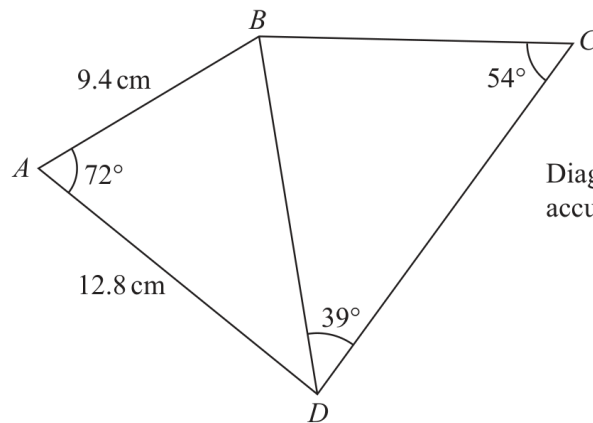
# FAMILY MAP

## Sine Cosine Rule Area of Triangles

|            |                                                        |                |
|------------|--------------------------------------------------------|----------------|
| <b>F01</b> | <b>Resolve mixed non-right-triangle chains</b>         | 6 Q   31 marks |
| <b>F02</b> | <b>Use the sine rule</b>                               | 3 Q   16 marks |
| <b>F03</b> | <b>Use the cosine rule</b>                             | 1 Q   3 marks  |
| <b>F04</b> | <b>Use area equals one-half <math>ab \sin C</math></b> | 1 Q   3 marks  |

**Working Unit 13**

4MA1-1H Q20

**UNIT 1**  
**5 marks****20**Diagram **NOT**  
accurately drawn

Work out the length of  $BC$   
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

**Question 13**

4MA1-1H Q19

MODULAR UNIT 1

5 marks

19 Here is quadrilateral  $ABCD$

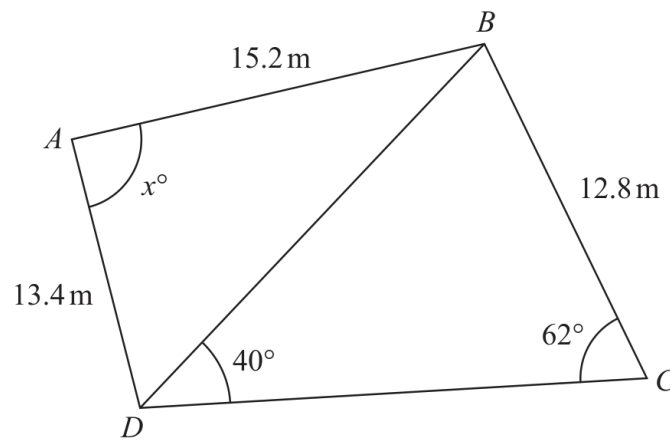


Diagram **NOT**  
accurately drawn

Work out the value of  $x$   
Give your answer correct to 3 significant figures.

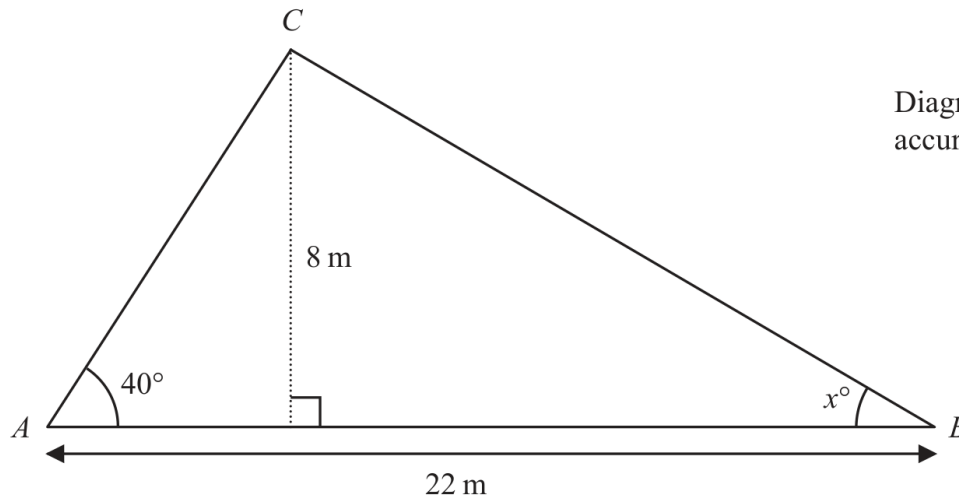
$x = \dots\dots\dots$

**Question 09**

4MA1-1H Q10 demand

MODULAR UNIT 1

5 marks

10 Here is triangle  $ABC$ 

Work out the value of  $x$   
Give your answer correct to one decimal place.  
Show your working clearly.

 $x = \dots\dots\dots$

**Question 26**

4MA1-2H Q22

MODULAR UNIT 1

5 marks

22 The diagram shows a triangle  $ABC$  and a flagpole  $BF$

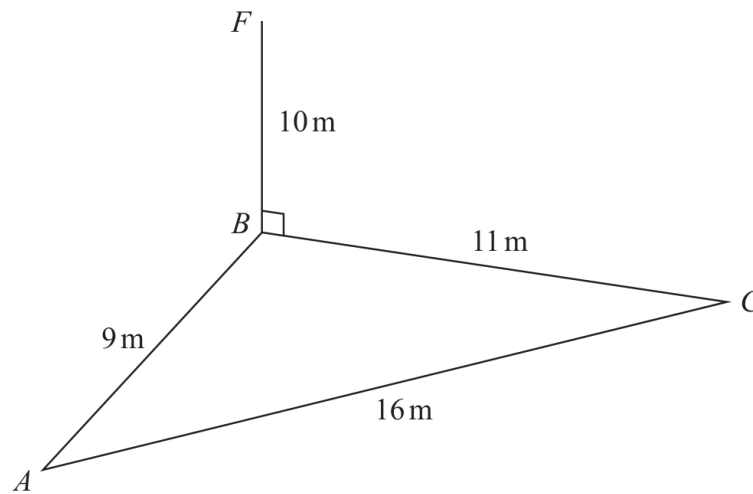


Diagram **NOT**  
accurately drawn

$A$ ,  $B$  and  $C$  are points on horizontal ground.

$BF$  is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

$D$  is the point on  $AC$  such that angle  $BDC = 90^\circ$

Work out the size of the angle of elevation of the point  $F$  from the point  $D$   
Give your answer correct to one decimal place.

**Question 12**

4MA1-1H Q18 Q18

MODULAR UNIT 1

5 marks

18 Here is triangle  $ABC$

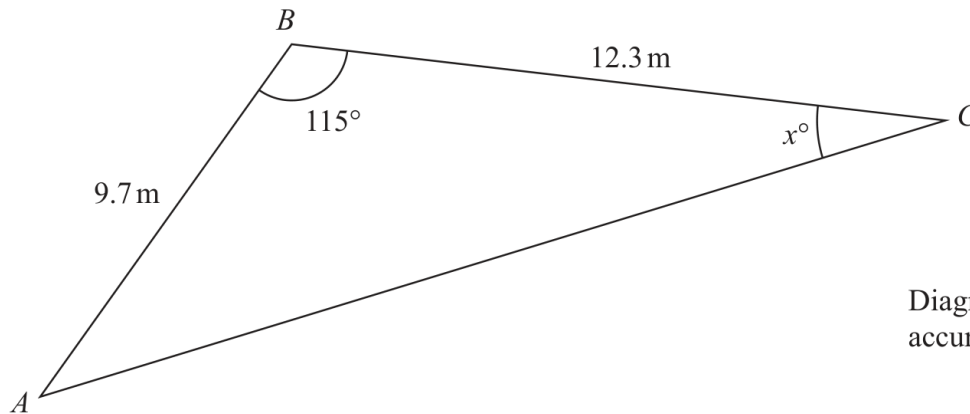


Diagram **NOT**  
accurately drawn

Work out the value of  $x$   
Give your answer correct to 3 significant figures.

**Question 27**

4MA1-2H Q23

MODULAR UNIT 1

6 marks

23 The diagram shows triangle  $PQR$

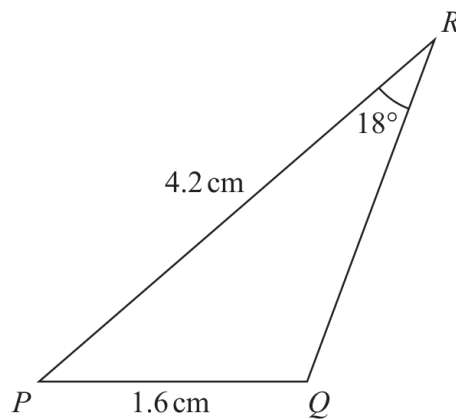


Diagram **NOT**  
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle  $PQR$  is obtuse,

work out the area of triangle  $PQR$

Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$

14 The diagram shows parallelogram  $EFGH$ .

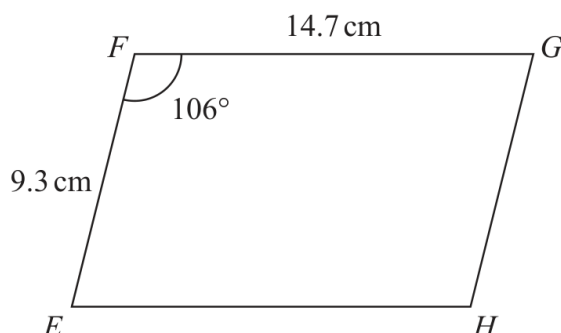


Diagram **NOT**  
accurately drawn

$$EF = 9.3 \text{ cm}$$

$$FG = 14.7 \text{ cm}$$

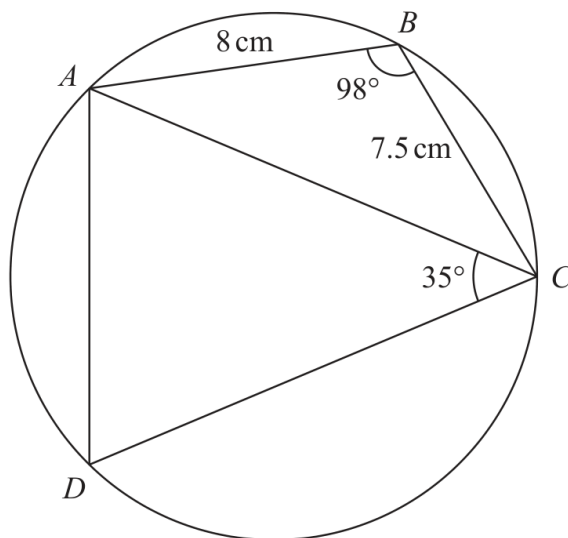
$$\text{Angle } EFG = 106^\circ$$

- (a) Work out the area of the parallelogram.  
Give your answer correct to 3 significant figures.

- (b) Work out the length of the diagonal  $EG$  of the parallelogram.  
Give your answer correct to 3 significant figures.

..... cm  
(3)

Diagram **NOT**  
accurately drawn


$$\begin{aligned} AB &= 8 \text{ cm} \\ BC &= 7.5 \text{ cm} \\ \text{Angle } ABC &= 98^\circ \\ \text{Angle } ACD &= 35^\circ \end{aligned}$$

..... cm

A quadrilateral  $ABCD$  is shown with diagonal  $AC$ . The side lengths are  $AB = 9\text{ cm}$ ,  $BC = 12\text{ cm}$ , and  $CD = 7\text{ cm}$ . The interior angles are  $\angle ABC = 60^\circ$  and  $\angle ACD = 84^\circ$ .

Calculate the area of quadrilateral  $ABCD$ .  
Give your answer correct to 3 significant figures.  
Show your working clearly.

..... cm<sup>2</sup>

The diagram shows two triangles,  $ABC$  and  $DEC$ , that intersect at vertex  $C$ . The vertices are labeled  $A$ ,  $B$ ,  $C$ ,  $D$ , and  $E$ . The side lengths are given as follows:  $BC = 31$  cm,  $AC = 24$  cm,  $EC = 19$  cm, and  $DC = 16$  cm. The angle at vertex  $A$ ,  $\angle BAC$ , is  $64^\circ$ .

 $AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$ 

Angle  $BAC = 64^\circ$

Work out the length of  $DE$

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 16 is 5 marks)

19 The diagram shows triangle  $ABC$

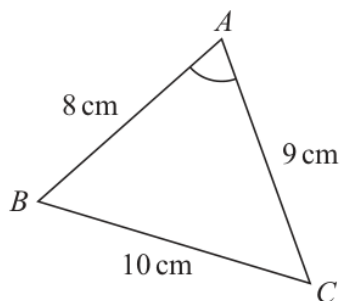


Diagram **NOT**  
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle  $BAC$

Give your answer correct to one decimal place.

(Total for Question 19 is 3 marks)